



SECP3723 SYSTEM DEVELOPMENT TECHNOLOGY

SESSION 2024/2025 – SEMESTER 1

INDIVIDUAL PROJECT

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Year / Program	2/SECPH
Section	02
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System	Course Registration System

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1 System Introduction

The system is a Course Registration System for students in a university. As the name suggests, it is a system for course registration. Students are able to register for courses, while lecturers are able to view the courses assigned and the students registered. Staff, on the other hand, can manipulate the courses for registration and the registrations made by students.

2 System Design

2.1 Use Case

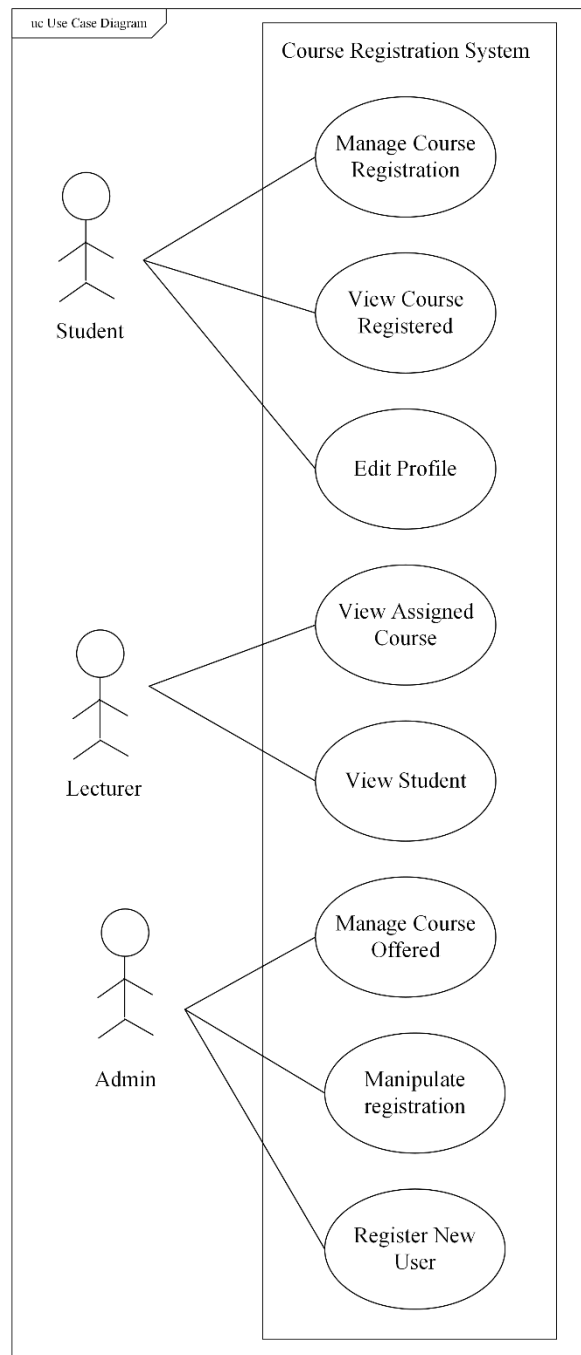


Figure 1 Use Case Diagram for Course Registration System

2.2 ERD

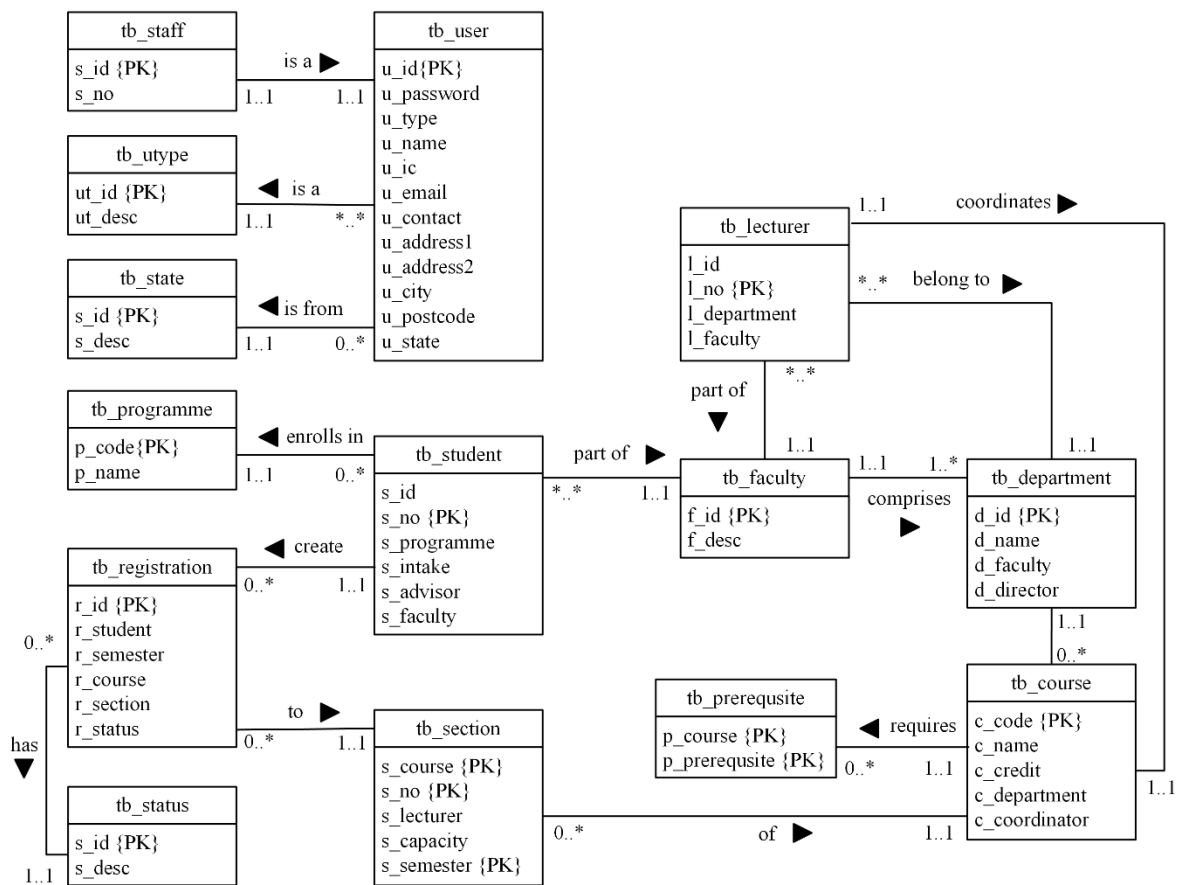


Figure 2 Entity Relationship Diagram for Course Registration System

3 Development Software, Language, Technology and Tools

The development software used are VSCode and XAMPP. VSCode is used to write and edit the code while XAMPP is the local server environment used to run and test the system.

- VSCode: <https://code.visualstudio.com/>
- XAMPP: <https://www.apachefriends.org/download.html>

The languages used to develop the system include HTML, CSS, SQL, PHP and JavaScript.

The technologies used include MySQL, Infinity Hosting, and AJAX. MySQL is the DBMS used to store the data, while Infinity Hosting is used to host the application for free online. AJAX can make requests without reloading, allowing dynamic content loading.

- Infinity Hosting: <https://www.infinityfree.com/>

Other tools include Selectize.js, SweetAlert, Font Awesome and Bootstrap. These tools greatly improve the User Experience by making the website more appealing and more easy to use.

Selectize.js allows users to type and search the selection without having to scroll through the list of options.

- Selectize.js (Website): <https://selectize.dev/>
- Selectize.js (Link): <https://cdnjs.cloudflare.com/ajax/libs/selectize.js/0.15.2/css/selectize.bootstrap5.min.css>

SweetAlert can substitute windows alert and make the alert more appealing.

- SweetAlert (Website): <https://sweetalert2.github.io/>
- SweetAlert (Link): <https://cdn.jsdelivr.net/npm/sweetalert2@11>

Font Awesome provides a library of icons to add to buttons.

- Font Awesome (Website): <https://fontawesome.com/>
- Font Awesome (Link): <https://cdnjs.cloudflare.com/ajax/libs/font-awesome/6.0.0-beta3/css/all.min.css>

Bootswatch provides basic styling and ensures consistency in design throughout the website. The "Litera" theme is used for the system.

- Bootswatch (Website): <https://bootswatch.com/>

4 Database Design

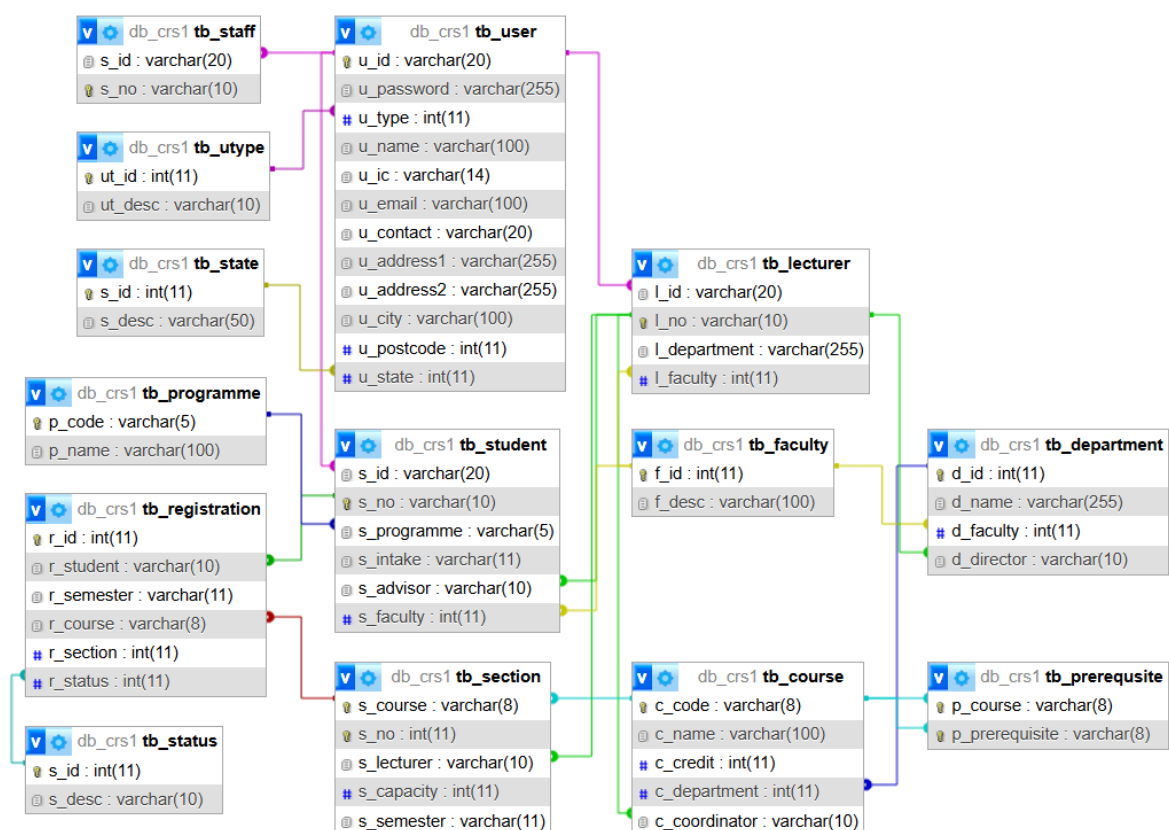


Figure 3 Database Structure and Relationship

5 Development Steps

System Analysis

A brief analysis is conducted to gather the requirements and design the database. The Use Case Diagram and Entity Relationship Diagram are drawn.

System Design and Development

The database is built based on the ERD. Some data is populated to test the system's functionality and structure. This ensures that queries will work as expected.

The development process focuses on implementing one feature at a time. Interface design and development are done simultaneously. Initially, a rough sketch of the interface is drawn on paper to visualize the layout. Throughout the coding process, the interface design evolves to accommodate the needs of the system or the limitations in the tools or technology.

As the coding progresses, new insights and ideas may arise, and the previous designs are refined and adjusted to improve user experience.

System Testing

As each unit or part is developed, testing is conducted to ensure that the system works correctly as expected. Any errors or issues found are resolved.

After parts of features are completed, testing is done across several related features to ensure functionality.

After all features are completed, testing is done again to ensure that all parts work together seamlessly.

After hosting, all features are tested again to ensure that there are no errors.

6 System Interface for All Users

Login

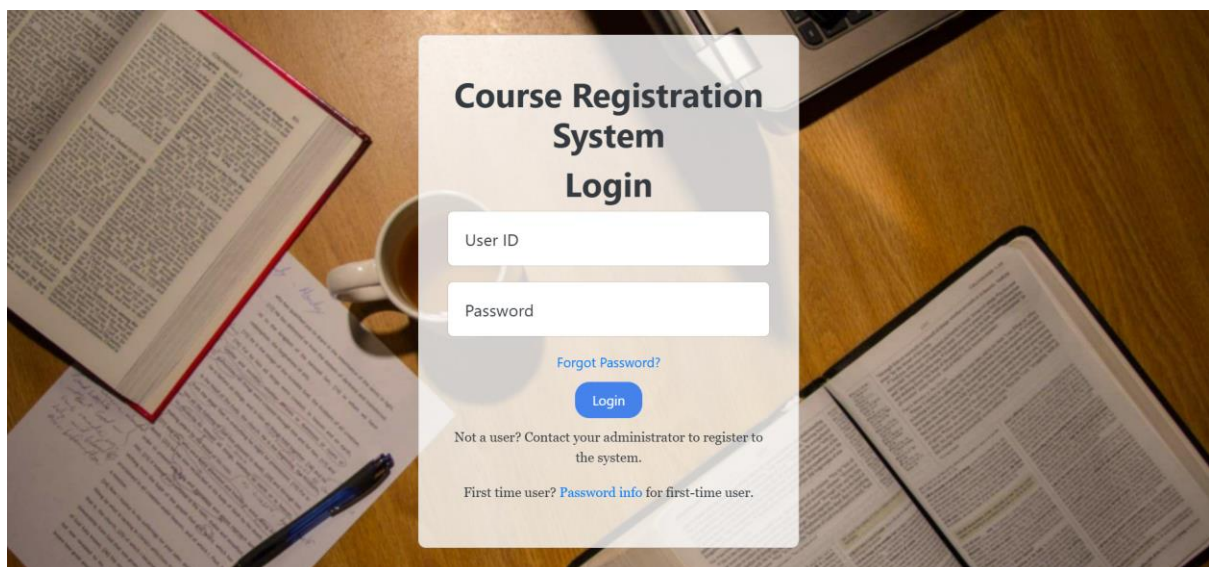


Figure 4 Login Page

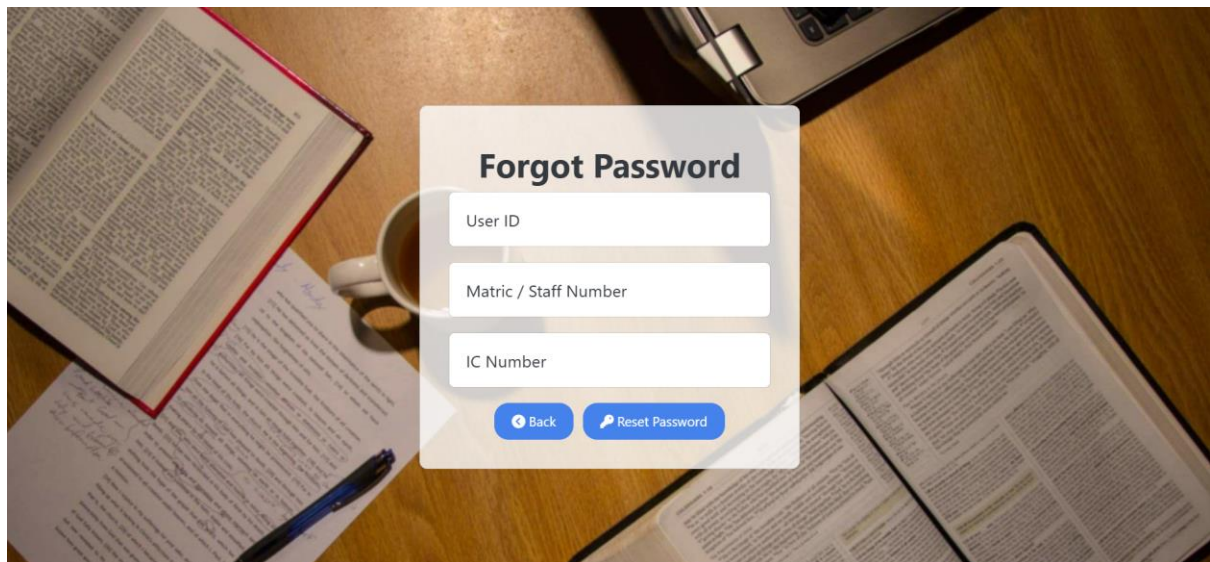


Figure 5 Forgot Password Page

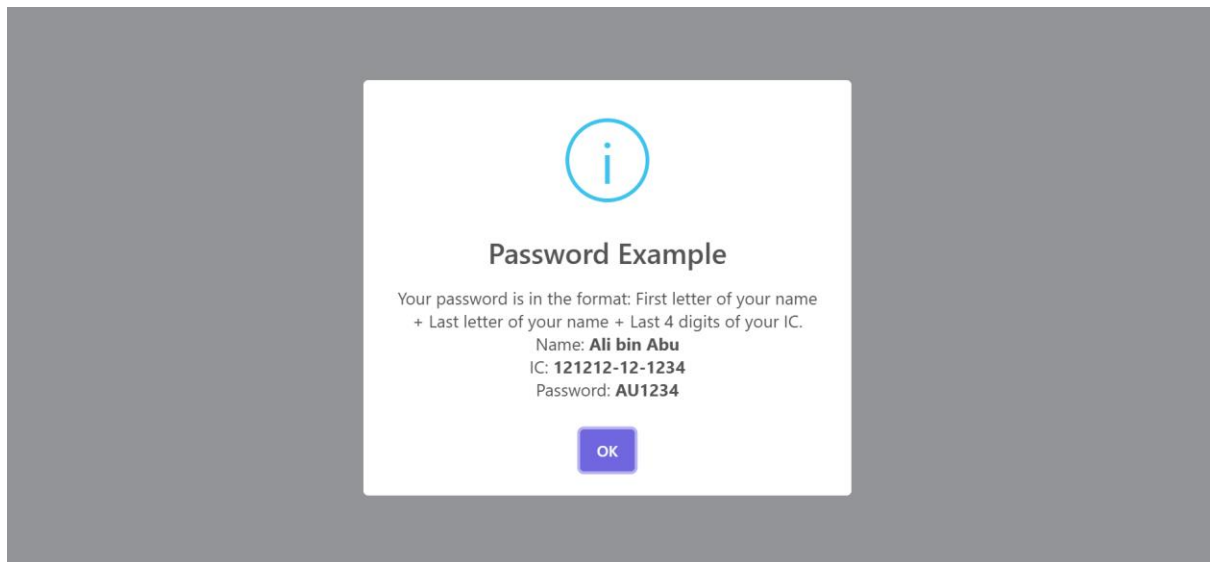


Figure 6 Password Info Page

6.1 Student

Course Registration

CoursesRegisterProfile

Logout

Hi, Lam Yoke Yu

Semester 2024/2025-1

Semester 2024/2025-1

Course Code	Course	Credit	Section	Lecturer
SECP2523	Database (WBL)	3	2	Dr. Noorfa Haszlinna binti Mustafa
SECJ1013	Programming Technique I	3	1	Assoc. Prof. Dr. Haza Nuzly bin Abdull Hamed
SECP3204	Software Engineering (WBL)	4	2	Dr. Mohd Fo'ad bin Rohani
SECP3723	System Development Technology (WBL)	3	1	Assoc. Prof. Dr. Haza Nuzly bin Abdull Hamed

Course Registration System Developed by Lam Yoke Yu @ 2025

Figure 7 Student Home Page

Course Registration

CoursesRegisterProfile

Logout

Registered Courses

Semester 2024/2025-1

Course Code	Course	Credit	Section	Lecturer
SECP2523	Database (WBL)	3	2	Dr. Noorfa Haszlinna binti Mustafa
SECJ1013	Programming Technique I	3	1	Assoc. Prof. Dr. Haza Nuzly bin Abdull Hamed
SECP3204	Software Engineering (WBL)	4	2	Dr. Mohd Fo'ad bin Rohani
SECP3723	System Development Technology (WBL)	3	1	Assoc. Prof. Dr. Haza Nuzly bin Abdull Hamed

Semester 2023/2024-2

Course Code	Course	Credit	Section	Lecturer
SECP3213	Business Intelligence	3	4	Ali bin Abu
SECJ1013	Discrete Structure	3	1	Assoc. Prof. Dr. Haza Nuzly bin Abdull Hamed
SECJ1143	Probability and Statistical Data Analysis	3	2	Dr. Muhammad Aliif bin Ahmad
SECP2613	System Analysis and Design (WBL)	3	1	Dr. Sharin Hazlin binti Huspi
SECP3723	System Development Technology (WBL)	3	1	Dr. Aryati binti Bakri

Course Registration System Developed by Lam Yoke Yu @ 2025

Figure 8 Student current and previous semester courses

Course Registration
[Courses](#)
[Register](#)
[Profile](#)
Logout

Course Registration

Browse Course

Course Code	Course	Section	Credit	Status	Action
SECP2523	Database (WBL)	2	3	Approved	Delete
SECJ1013	Programming Technique I	1	3	Approved	Delete
SECP3204	Software Engineering (WBL)	2	4	Approved	Delete
SECP3723	System Development Technology (WBL)	1	3	Approved	Delete

Select Course

Select Section

New

Delete

+ Add Course

Save

Course Registration System Developed by Lam Yoke Yu @ 2025

Figure 9 Student Register Course and Cancel Registration Page

Course Registration
[Courses](#)
[Register](#)
[Profile](#)
Logout

Course List

SECI1013 Discrete Structure

SECI1143 Probability and Statistical Data Analysis

Credit Hour: 3

Section	Capacity	Action
1	0/0	Register
2	0/0	Register

SECJ1013 Programming Technique I Registered

SECJ2013 Data Structure and Algorithm

SECP2523 Database (WBL) Registered

SECP3204 Software Engineering (WBL) Registered

SECP3723 System Development Technology (WBL) Registered

SECP3744 Enterprise Systems Design and Modeling (WBL)

SECR2213 Network Communication

1

Course Registration System Developed by Lam Yoke Yu @ 2025

Figure 10 Student Search and register course page

Course Registration

CoursesRegisterProfile

Logout

Profile

Name

Lam Yoke Yu

Matric Number

A23CS0233

IC Number

000000-00-0000

Email address

lam.yu@graduate.utm.my

Contact Number

0123456789

Address

Jalan Rishah

Taman Rishah

City

Ipoh

State

Perak

Postcode

30100

Reset Password

Save

Course Registration System Developed by Lam Yoke Yu @ 2025

Figure 11 Student Edit Profile Page

Course Registration

CoursesRegisterProfile

Logout

Reset Password

Current Password

New Password

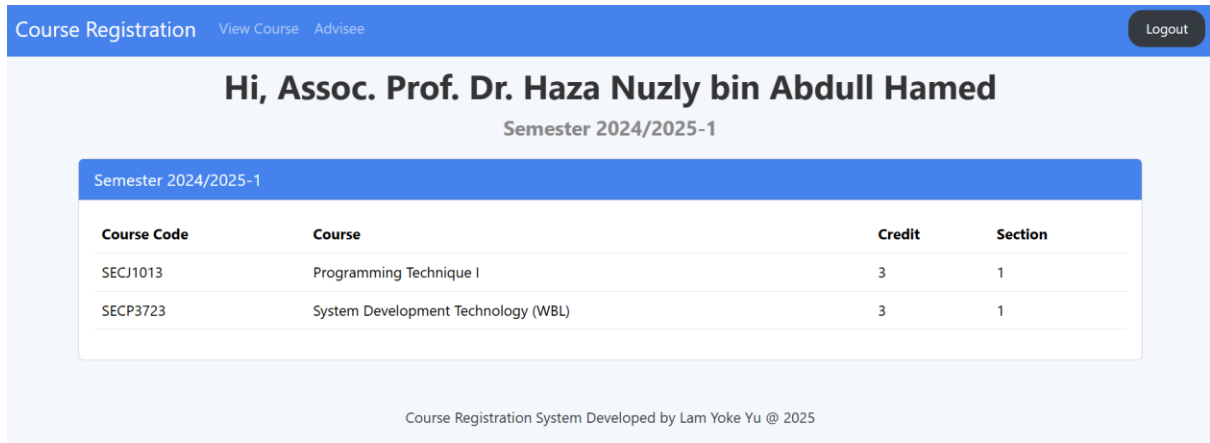
Confirm Password

Reset Password

Course Registration System Developed by Lam Yoke Yu @ 2025

Figure 12 Student Reset Password Page

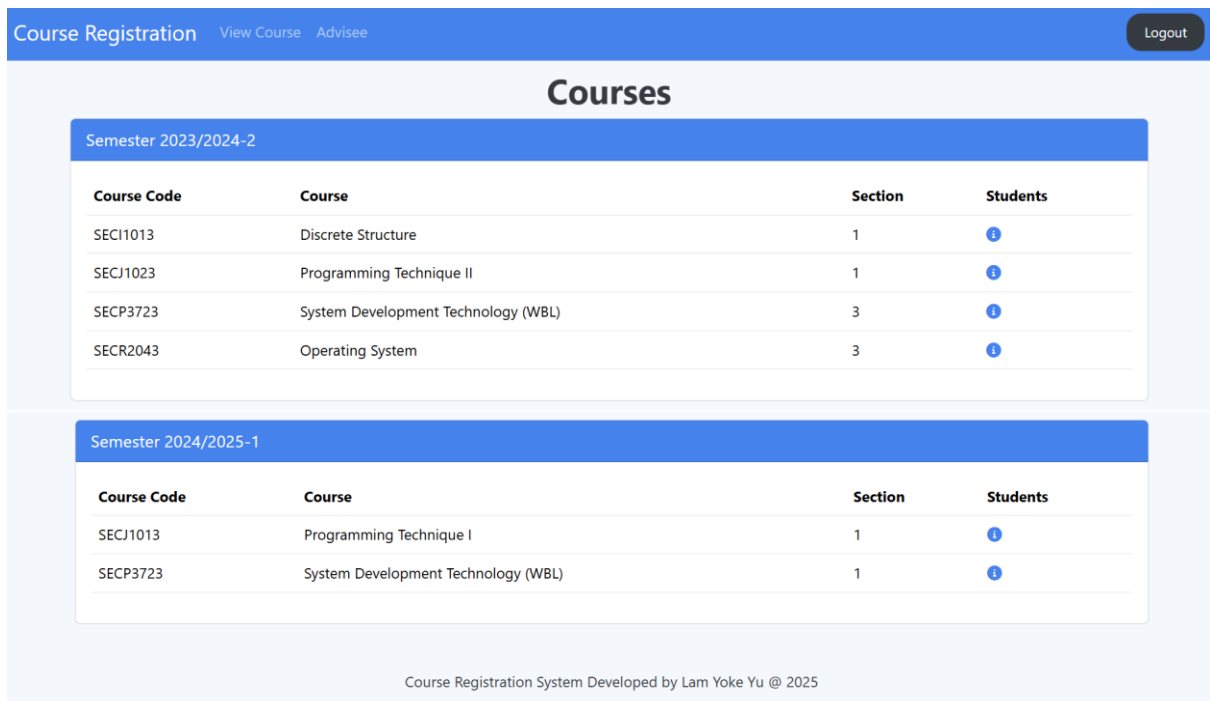
6.2 Lecturer



The screenshot shows the 'Lecturer Home Page' of a 'Course Registration' system. The header bar is blue with 'Course Registration' on the left, 'View Course' and 'Advisee' in the center, and a 'Logout' button on the right. The main content area has a light blue background. At the top, it says 'Hi, Assoc. Prof. Dr. Haza Nuzly bin Abdull Hamed' followed by 'Semester 2024/2025-1'. Below this is a table for 'Semester 2024/2025-1' with columns: Course Code, Course, Credit, and Section. The table lists two courses: SECJ1013 (Programming Technique I, 3 credits, 1 section) and SECP3723 (System Development Technology (WBL), 3 credits, 1 section). At the bottom, it says 'Course Registration System Developed by Lam Yoke Yu @ 2025'.

Course Code	Course	Credit	Section
SECJ1013	Programming Technique I	3	1
SECP3723	System Development Technology (WBL)	3	1

Figure 13 Lecturer Home Page



The screenshot shows the 'View assigned courses for each semester' page. The header bar is blue with 'Course Registration' on the left, 'View Course' and 'Advisee' in the center, and a 'Logout' button on the right. The main content area has a light blue background. At the top, it says 'Courses'. Below this is a table for 'Semester 2023/2024-2' with columns: Course Code, Course, Section, and Students. The table lists four courses: SECJ1013 (Discrete Structure, 1 section, 1 student), SECJ1023 (Programming Technique II, 1 section, 1 student), SECP3723 (System Development Technology (WBL), 3 sections, 1 student), and SECR2043 (Operating System, 3 sections, 1 student). Below this is another table for 'Semester 2024/2025-1' with columns: Course Code, Course, Section, and Students. The table lists two courses: SECJ1013 (Programming Technique I, 1 section, 1 student) and SECP3723 (System Development Technology (WBL), 1 section, 1 student). At the bottom, it says 'Course Registration System Developed by Lam Yoke Yu @ 2025'.

Course Code	Course	Section	Students
SECJ1013	Discrete Structure	1	1
SECJ1023	Programming Technique II	1	1
SECP3723	System Development Technology (WBL)	3	1
SECR2043	Operating System	3	1

Course Code	Course	Section	Students
SECJ1013	Programming Technique I	1	1
SECP3723	System Development Technology (WBL)	1	1

Figure 14 View assigned courses for each semester

Course Registration
View Course
Advisee
Logout

Students

SECI1013 Discrete Structure

Credit	3
Coordinator	1112 Ts. Dr. Johanna binti Ahmad
Department	Emergent Computing

Student Name	Matric Number	Intake
Dayang Farah	A23CS0002	2024/2025-1
Lam Yoke Yu	A23CS0233	2023/2024-1
Tan Yi Ya	A23CS0187	2023/2024-1

Course Registration System Developed by Lam Yoke Yu @ 2025

Figure 15 View course details and student details

Course Registration
View Course
Advisee
Logout

Advisee

Roslan bin Roslan A23CS0001

Course Code	Course	Section	Credit	Status	Action
SECP2523	Database (WBL)	2	3	Approved	No Action

Dayang Farah A23CS0002

Course Code	Course	Section	Credit	Status	Action
SECP2523	Database (WBL)	2	3	Approved	No Action

Teh Ru Qian A23CS0191

Course Code	Course	Section	Credit	Status	Action
SECI1013	Programming Technique I	1	3	Approved	No Action
SECP2523	Database (WBL)	2	3	Approved	No Action

Lam Yoke Yu A23CS0233

Course Code	Course	Section	Credit	Status	Action
SECP2523	Database (WBL)	2	3	Approved	No Action
SECI1013	Programming Technique I	1	3	Approved	No Action
SECP3204	Software Engineering (WBL)	2	4	Approved	No Action
SECI2013	Data Structure and Algorithm	2	3	Approved	No Action

Course Registration System Developed by Lam Yoke Yu @ 2025

Figure 16 View Advisee Course Registration Status

6.3 Admin



Figure 17 Admin Home Page

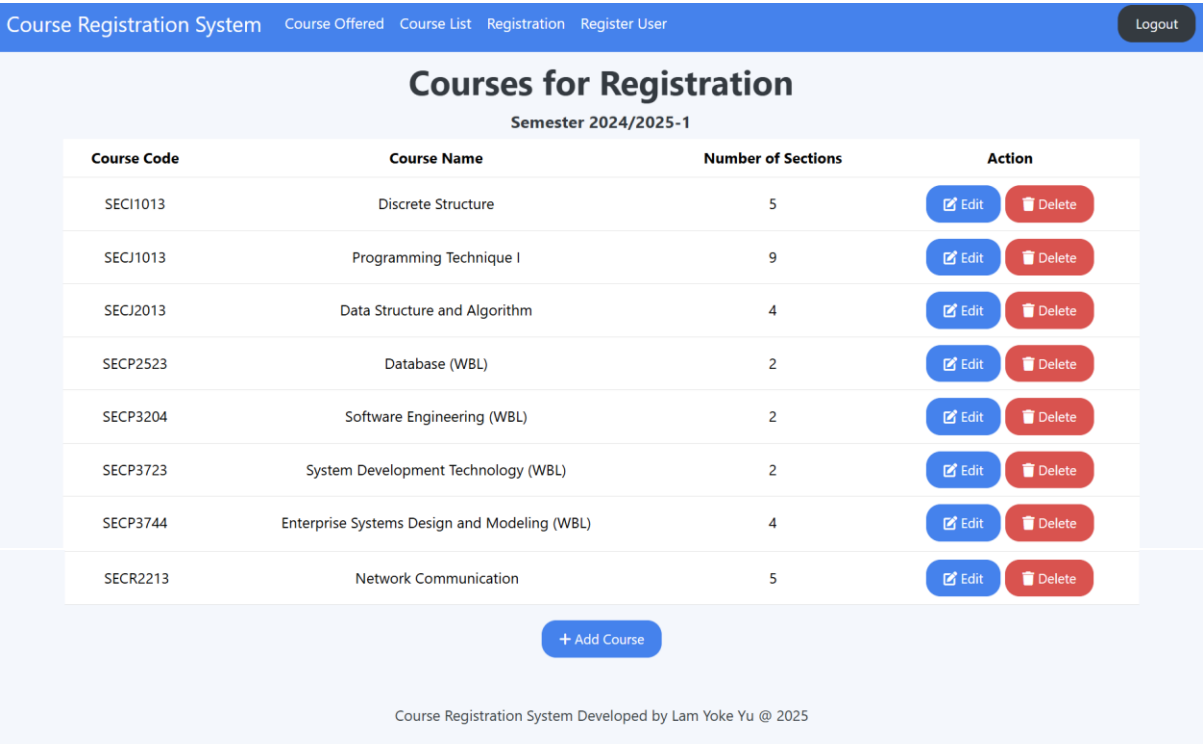


Figure 18 Modify List of Course Offer for the Semester

Course Registration System
Course Offered
Course List
Registration
Register User
Logout

Add Courses for Registration

Course Code
Choose Course

Number of Sections

+ Add Sections

Course Registration System Developed by Lam Yoke Yu @ 2025

Figure 19 Page After Clicking “Add Course”

Course Registration System
Course Offered
Course List
Registration
Register User
Logout

Edit Sections

SECP3723 - System Development Technology (WBL)
Credit: 3
Coordinator: 10199 - Assoc. Prof. Dr. Haza Nuzly bin Abdull Hamed
Department: Emergent Computing

Section	Lecturer	Capacity
1	10199 - Assoc. Prof. Dr. Haza Nuzly bin Abdull Hamed	30
2	1116 - Dr. Noorfa Haszlinna binti Mustaffa	30
3	Choose Lecturer	0
4	Choose Lecturer	0

Back
Save

Course Registration System Developed by Lam Yoke Yu @ 2025

Figure 20 Page After Entering Details and Clicking “Add Sections”

Course Registration System
Course Offered
Course List
Registration
Register User
Logout

Course List

Select Department
Search

Course Code	Course	Credit	Department	Coordinator	Action
SECD3761	Technopreneurship Seminar (WBL)	1	Emergent Computing	Dr. Mohd Fo'ad bin Rohani	Edit
SECI1013	Discrete Structure	3	Emergent Computing	Ts. Dr. Johanna binti Ahmad	Edit
SECI1113	Computational Mathematics	3	Software Engineering	Dr. Mohd Fo'ad bin Rohani	Edit
SECI1143	Probability and Statistical Data Analysis	3	Applied Computing And Artificial Intelligence	Dr. Muhammad Aliif bin Ahmad	Edit
SECJ1013	Programming Technique I	3	Emergent Computing	Dr. Muhammad Iqbal Tariq bin Idris	Edit
SECJ1023	Programming Technique II	3	Software Engineering	Dr. Noorfa Haszlinna binti Mustaffa	Edit
SECJ2013	Data Structure and Algorithm	3	Computer Science	Dr. Mohd Razak bin Samingan	Edit
SECJ2154	Object-Oriented Programming	4	Software Engineering	Dr. Nur Eiliyah @ Wong Yee Leng	Edit
SECJ2253	Requirements Engineering & Software Modelling	3	Applied Computing And Artificial Intelligence	Rozilawati binti Dollah @ Md Zain	Edit
SECJ2363	Software Project Management	3	Computer Science	Dr. Zuraini binti Ali Shah	Edit

+ Add Course

1 2 3 4 5 »

Figure 21 Course List

Course Registration System
Course Offered
Course List
Registration
Register User
Logout

Edit Course

Course Code
SECJ1023

Course Name
Programming Technique II

Credit
3

Department
Software Engineering

Coordinator

Dr. Noorfa Haszlinna binti Mustafa

Prerequisites

Programming Technique I (SECJ1013)

Add Another Prerequisite

Back Save

Course Registration System Developed by Lam Yoke Yu @ 2025

Figure 22 Edit A Course

Course Registration System Course Offered Course List Registration Register User Logout

Add Course

Course Code

Course Code

Course Name

Course Name

Credit

Credit

Department

Select Department

Coordinator

Select Lecturer

Back Save

Course Registration System Developed by Lam Yoke Yu @ 2025

Figure 23 Page For Adding New Course

Course Registration System						
Course Offered Course List Registration Register User Logout						
Registration						
Semester 2024/2025-1						
Select Course Section Search Search						
Student	Name	Course Code	Course	Section	Status	Action
A23CS0233	Lam Yoke Yu	SECP2523	Database (WBL)	2	Approved	Edit
A23CS0240	Lim Xin Rou	SECP2523	Database (WBL)	1	Approved	Edit
A23CS0240	Lim Xin Rou	SECP3723	System Development Technology (WBL)	2	Approved	Edit
A23CS0233	Lam Yoke Yu	SECJ1013	Programming Technique I	1	Approved	Edit
A23CS0187	Tan Yi Ya	SECJ1013	Programming Technique I	1	Approved	Edit
A23CS0191	Teh Ru Qian	SECJ1013	Programming Technique I	1	Approved	Edit
A23CS0191	Teh Ru Qian	SECP2523	Database (WBL)	2	Approved	Edit
A23CS0233	Lam Yoke Yu	SECP3204	Software Engineering (WBL)	2	Approved	Edit
A23CS0233	Lam Yoke Yu	SECJ2013	Data Structure and Algorithm	2	Approved	Edit
A23CS0240	Lim Xin Rou	SECJ2013	Data Structure and Algorithm	3	Approved	Edit
12»						
Course Registration System Developed by Lam Yoke Yu @ 2025						

Figure 24 List of Registration with Filters

Course Registration System		Course Offered	Course List	Registration	Register User	Logout
Edit Registration						
Student Information						Student Registration
Name	Lam Yoke Yu					
Matric Number	A23CS0233					
Programme Code	SECPH					
Intake	2023/2024-1					
Advisor	Assoc. Prof. Dr. Haza Nuzly bin Abdull Hamed					

Section Information

Section Registration

Course Code	SECP2523
Course Name	Database (WBL)
Section No	2
Lecturer	Dr. Noorfa Haszlinna binti Mustaffa

Approved

Back

Save

Course Registration System Developed by Lam Yoke Yu @ 2025

Figure 25 Edit A Registration

Course Registration System

Course Offered

Course List

Registration

Register User

Logout

Student Registration

Student Information

Name	Lam Yoke Yu
Matric Number	A23CS0233
Programme Code	SECPH
Intake	2023/2024-1
Advisor	Assoc. Prof. Dr. Haza Nuzly bin Abdull Hamed

Course Code	Course	Section	Credit	Status	Action
SECP2523	Database (WBL)	2	3	Approved	Delete
SECJ1013	Programming Technique I	1	3	Approved	Delete
SECP3204	Software Engineering (WBL)	2	4	Approved	Delete
SECJ2013	Data Structure and Algorithm	2	3	Approved	Delete
Select Course		Select Section		New	Delete

+ Add Course

Save

Course Registration System Developed by Lam Yoke Yu @ 2025

Figure 26 Amend Specific Student Registration After Clicking “Student Registration”

Course Registration System
Course Offered
Course List
Registration
Register User
Logout

Section Registration

Section Information

Lecturer	1116 - Dr. Noorfa Haszlinna binti Mustaffa
Capacity	6/30
Credit	3
Coordinator	1123 - Assoc. Prof. Ts. Dr. Farhan bin Mohamed
Department	Computer Science

Matric Number	Name	Intake	Programme	Action
A23CS0233	Lam Yoke Yu	2023/2024-1	SECPH	Delete
A23CS0191	Teh Ru Qian	2023/2024-1	SECPH	Delete
A23CS0002	Dayang Farah	2024/2025-1	SECPH	Delete
A23CS0001	Roslan bin Roslan	2024/2025-1	SECJH	Delete
A23CS0187	Tan Yi Ya	2023/2024-1	SECPH	Delete
A23CS0224	Goe Jie Ying	2023/2024-1	SECPH	Delete

+ Add Student
Save

Course Registration System Developed by Lam Yoke Yu @ 2025

Figure 27 Amend Specific Student Registration After Clicking “Section Registration”

Course Registration System
Course Offered
Course List
Registration
Register User
Logout

Register New User

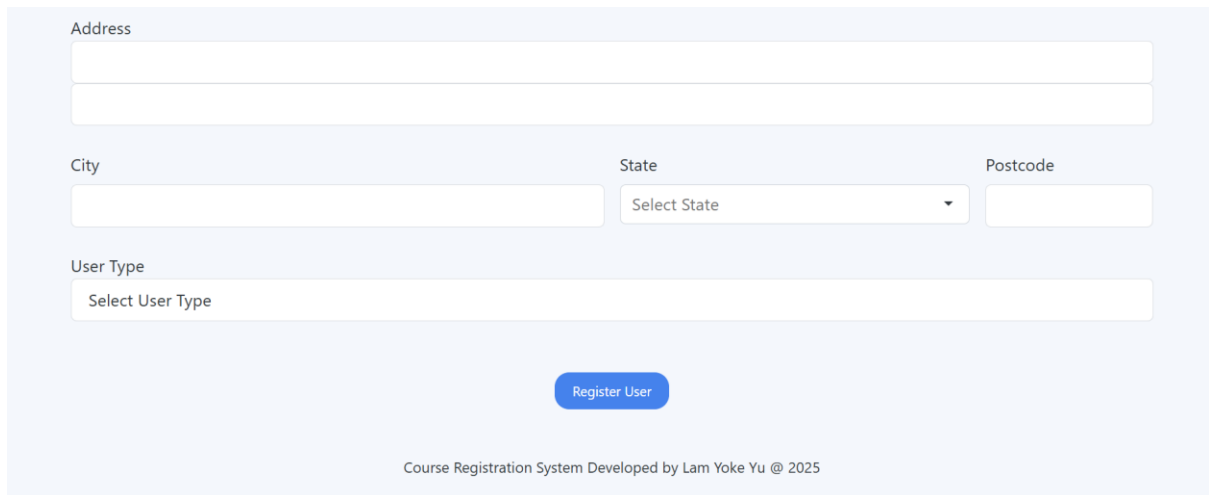
Full Name

User ID

Email

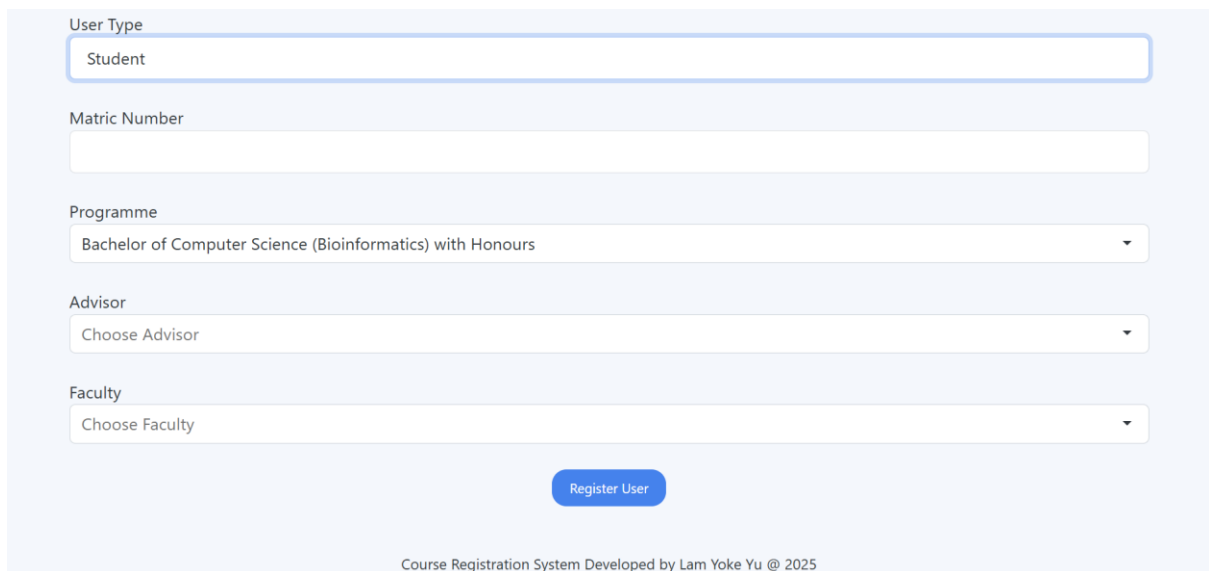
IC Number

Contact Number



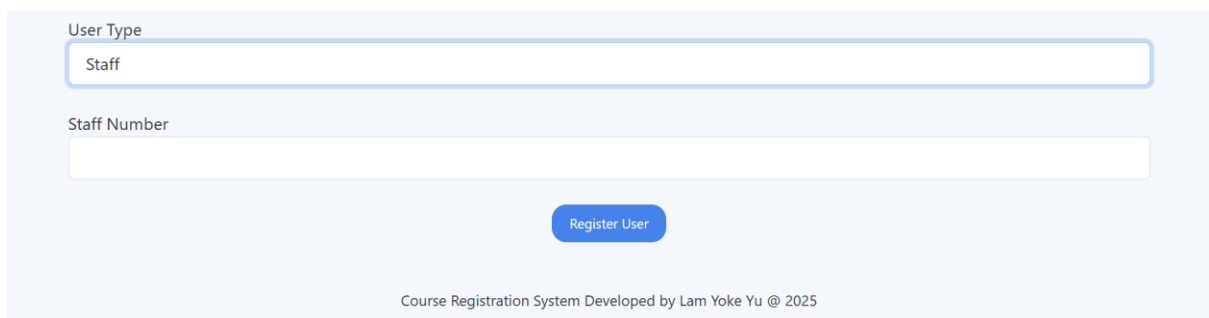
A registration form for a course registration system. It includes fields for Address (two lines), City, State (a dropdown menu showing 'Select State'), Postcode, User Type (a dropdown menu showing 'Select User Type'), and a 'Register User' button. The footer text reads 'Course Registration System Developed by Lam Yoke Yu @ 2025'.

Figure 28 Register New User



The registration form with 'Student' selected in the User Type dropdown. The form now includes fields for Matric Number, Programme (a dropdown menu showing 'Bachelor of Computer Science (Bioinformatics) with Honours'), Advisor (a dropdown menu showing 'Choose Advisor'), and Faculty (a dropdown menu showing 'Choose Faculty'). The 'Register User' button and footer text remain the same.

Figure 29 Form changes as the user type student is selected



The registration form with 'Staff' selected in the User Type dropdown. The form now includes a field for Staff Number. The 'Register User' button and footer text remain the same.

Figure 30 Form changes as the user type staff is selected

User Type
Lecturer

Staff Number

Department
Choose Department

Faculty
Choose Faculty

Register User

Course Registration System Developed by Lam Yoke Yu @ 2025

Figure 31 Form changes as the user type lecturer is selected

7 Localhost Setup

XAMPP is used to set up a local server environment to run and test the system. In the htdocs folder, a folder named crs is created to store files for the web application. The database is configured in db_connect.php, which is included in each page later. The system can then be run and tested using localhost/crs. The database is accessed using phpMyAdmin to track changes.

8 Hosting URL

<http://lamyokeyu.wuaze.com/>

9 User Credentials

User Type	User ID	Password	Information for Forgot Password		
			Matric / Staff Number	IC Number	Full Name
Student	lam.yu	12345678	A23CS0233	000000-00-0000	Lam Yoke Yu
	roslan	RN0000	A23CS0001	000000-00-0000	Roslan bin Roslan
Lecturer	haza	AD0000	10199	000000-00-0000	Assoc. Prof. Dr. Haza Nuzly bin Abdull Hamed
	hakim	HM0000	1129	000000-00-0000	Hakim bin Hakim
Staff	mawarabdul	ML0000	2002	000000-00-0000	Mawar bin Abdul